

Review Lecture (post Lecture 6)

September 14, 2023

Sample

Suppose that a week before the 2020 presidential election, you contacted a sample of $n = 625$ potential Georgia voters, randomly selected (with replacement) from the population of $N = 7,233,584$ on the public voters register, to ask whether they planned to vote.

Sample

Suppose that a week before the 2020 presidential election, you contacted a sample of $n = 625$ potential Georgia voters, randomly selected (with replacement) from the population of $N = 7,233,584$ on the public voters register, to ask whether they planned to vote.

Suppose also that all potential voters can be contacted, will respond honestly to your questions, and will not change their minds about voting.

Sample

Suppose that a week before the 2020 presidential election, you contacted a sample of $n = 625$ potential Georgia voters, randomly selected (with replacement) from the population of $N = 7,233,584$ on the public voters register, to ask whether they planned to vote.

Suppose also that all potential voters can be contacted, will respond honestly to your questions, and will not change their minds about voting.

Table: Sample

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.66

Sample

Suppose that a week before the 2020 presidential election, you contacted a sample of $n = 625$ potential Georgia voters, randomly selected (with replacement) from the population of $N = 7,233,584$ on the public voters register, to ask whether they planned to vote.

Suppose also that all potential voters can be contacted, will respond honestly to your questions, and will not change their minds about voting.

Table: Sample

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.66

Remember that the mean of a binary variable is a proportion.

Sample versus Population

Table: Sample

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.68

Sample versus Population

Table: Sample

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.68

After election day, we found that in fact 4, 999, 960 people, or roughly 70% of registered voters, turned out to vote.

Table: Population

j	1	2	3	4	...	...	...	...	7,233,584	$\bar{y}_{7,233,584}$
y_j	0	1	0	1	...	...	...	...	1	.70

Sampling Distribution of $\bar{Y}_{625}$

i	J_1	Y_1	J_2	Y_2	$\dots$	J_{625}	Y_{625}	$\bar{Y}_{625}$
	9562350	1	8763351	1	$\dots$	1294801	0	.68

Sampling Distribution of $\bar{Y}_{625}$

i	1		2		...	625		
s	J_1	Y_1	J_2	Y_2	...	J_{625}	Y_{625}	$\bar{Y}_{625}$
1	9562350	1	8763351	1	...	1294801	0	.68
2	5331704	0	4533839	1	...	3342359	1	.70

Sampling Distribution of $\bar{Y}_{625}$

i	1		2		...	625		
s	J_1	Y_1	J_2	Y_2	...	J_{625}	Y_{625}	$\bar{Y}_{625}$
1	9562350	1	8763351	1	...	1294801	0	.68
2	5331704	0	4533839	1	...	3342359	1	.70
3	5129936	0	10981600	0	...	4096184	1	.75
4	803605	0	7036389	1	...	803605	0	.73
5	148567	0	3833847	1	...	4769869	1	.69
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	4163458	0	8384613	1	...	377981	1	.74
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Activity 1

- ▶ Draw the pmf for Y_1 (or any other sampled observation).

Activity 1

- ▶ Draw the pmf for Y_1 (or any other sampled observation).
- ▶ Draw $E[Y_1]$ on that plot as a fulcrum.

Activity 1

- ▶ Draw the pmf for Y_1 (or any other sampled observation).
- ▶ Draw $E[Y_1]$ on that plot as a fulcrum.
- ▶ Sketch out the sampling distribution for the sample mean/proportion based on $n=21$.

Activity 1

- ▶ Draw the pmf for Y_1 (or any other sampled observation).
- ▶ Draw $E[Y_1]$ on that plot as a fulcrum.
- ▶ Sketch out the sampling distribution for the sample mean/proportion based on $n=21$.
- ▶ What are the mean and standard deviation of that sampling distribution you drew?

Activity 1

- ▶ Draw the pmf for Y_1 (or any other sampled observation).
- ▶ Draw $E[Y_1]$ on that plot as a fulcrum.
- ▶ Sketch out the sampling distribution for the sample mean/proportion based on $n=21$.
- ▶ What are the mean and standard deviation of that sampling distribution you drew?
- ▶ What do we call the standard deviation of a sampling distribution?

Activity 1

- ▶ Draw the pmf for Y_1 (or any other sampled observation).
- ▶ Draw $E[Y_1]$ on that plot as a fulcrum.
- ▶ Sketch out the sampling distribution for the sample mean/proportion based on $n=21$.
- ▶ What are the mean and standard deviation of that sampling distribution you drew?
- ▶ What do we call the standard deviation of a sampling distribution?
- ▶ What is the shape of the sampling distribution you drew?

Activity 2

- ▶ Draw the bootstrapped sampling distribution on the same plot.

Activity 2

- ▶ Draw the bootstrapped sampling distribution on the same plot.
- ▶ What is the mean of this distribution?

Activity 2

- ▶ Draw the bootstrapped sampling distribution on the same plot.
- ▶ What is the mean of this distribution?
- ▶ Is the standard deviation of this distribution larger or smaller than the standard deviation of the sampling distribution?

Activity 2

- ▶ Draw the bootstrapped sampling distribution on the same plot.
- ▶ What is the mean of this distribution?
- ▶ Is the standard deviation of this distribution larger or smaller than the standard deviation of the sampling distribution?
- ▶ What do we call this standard deviation?

Activity 2

- ▶ Draw the bootstrapped sampling distribution on the same plot.
- ▶ What is the mean of this distribution?
- ▶ Is the standard deviation of this distribution larger or smaller than the standard deviation of the sampling distribution?
- ▶ What do we call this standard deviation?
- ▶ How would you construct a 95% CI from this distribution?

Activity 2

- ▶ Draw the bootstrapped sampling distribution on the same plot.
- ▶ What is the mean of this distribution?
- ▶ Is the standard deviation of this distribution larger or smaller than the standard deviation of the sampling distribution?
- ▶ What do we call this standard deviation?
- ▶ How would you construct a 95% CI from this distribution?
- ▶ If I tell you that the estimated standard error is 0.1, construct a 95% CI using the normal approximation formula.

Sampling Distribution for the Sample Mean

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
$f.(.)$	?	?	...	?	?
$E.(.)$	μ	μ	...	μ	?
$V.(.)$	σ^2	σ^2	...	σ^2	?

Sampling Distribution for the Sample Mean

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f.(.)$	?	?	...	?	?
$E.(.)$	μ	μ	...	μ	?
$V.(.)$	σ^2	σ^2	...	σ^2	?

Activity 3

- ▶ What are the expectation and variance of $\bar{Y}$?

Activity 3

- ▶ What are the expectation and variance of $\bar{Y}$?
- ▶ Suppose we somehow knew that $\mu = 130,000$ and $\sigma = 10,000$, draw your best guess for the pdf of Y_1 (or any other observation) and draw the sampling distribution of an $n = 100$ sample mean.

Activity 3

- ▶ What are the expectation and variance of $\bar{Y}$?
- ▶ Suppose we somehow knew that $\mu = 130,000$ and $\sigma = 10,000$, draw your best guess for the pdf of Y_1 (or any other observation) and draw the sampling distribution of an $n = 100$ sample mean.
- ▶ What is the mean of the sampling distribution and what is the standard deviation of the sampling distribution?

Activity 3

- ▶ What are the expectation and variance of $\bar{Y}$?
- ▶ Suppose we somehow knew that $\mu = 130,000$ and $\sigma = 10,000$, draw your best guess for the pdf of Y_1 (or any other observation) and draw the sampling distribution of an $n = 100$ sample mean.
- ▶ What is the mean of the sampling distribution and what is the standard deviation of the sampling distribution?
- ▶ Suppose the sample mean is $\bar{y} = 135,000$ and the sample standard deviation is $s = 8,100$, what is the estimated standard error?

Activity 3

- ▶ What are the expectation and variance of $\bar{Y}$?
- ▶ Suppose we somehow knew that $\mu = 130,000$ and $\sigma = 10,000$, draw your best guess for the pdf of Y_1 (or any other observation) and draw the sampling distribution of an $n = 100$ sample mean.
- ▶ What is the mean of the sampling distribution and what is the standard deviation of the sampling distribution?
- ▶ Suppose the sample mean is $\bar{y} = 135,000$ and the sample standard deviation is $s = 8,100$, what is the estimated standard error?
- ▶ using the normal approximation formula, calculate a 95% CI for μ ?

Activity 4

- ▶ Suppose we propose a class of estimators $\hat{\mu} = \sum_{i=1} c_i Y_i$, what must be true for $\hat{\mu}$ to be unbiased?

Activity 4

- ▶ Suppose we propose a class of estimators $\hat{\mu} = \sum_{i=1} c_i Y_i$, what must be true for $\hat{\mu}$ to be unbiased?
- ▶ What is the variance of $\hat{\mu}$?

Activity 5

Suppose the 4-year degree subpopulation mean is μ_1 and subpopulation variance is σ_1^2 , the non 4-year degree subpopulation mean is μ_0 and subpopulation variance is σ_0^2 . Suppose also that we have a stratified random sample with n_1 from the 4-year degree subpopulation and n_0 from the non 4-year degree subpopulation.

- ▶ Prove that $\bar{y}_1 - \bar{y}_0$ is unbiased for $\mu_1 - \mu_0$.
- ▶ Calculate the variance of $\bar{y}_1 - \bar{y}_0$.